Charger

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***Abstract*—In this report, we use a full-wave-rectified signal to review periodic signals, Fourier series, Fourier transforms, and filtering. We then describe the main stages of a regulated DC supply and present our measured USB output values.**

1. Periodic Function

1.1. Consider the full-wave-rectified sinusoidal signal

$$x(t) = A_0 \left|\sin\left(2\pi f_0 t\right)\right|, \tag{1.1}$$

Here, A_0 is the peak amplitude and f_0 is the frequency of the original sinusoid. Taking the absolute value turns both half-cycles positive, so the signal repeats twice during each cycle of the original sinusoid.

1.1 Plot $x(t)$.

1.2 Show that $x(t)$ is periodic and find its period. In particular, distinguish the period of $x(t)$ from the period of the underlying sinusoid.

2. Fourier Series

For our numerical calculations, we use $A_0 = 12\,\mathrm{V}$ and $f_0 = 50\,\mathrm{Hz}$. The rectified signal therefore has a fundamental frequency of $2f_0 = 100\,\mathrm{Hz}$.

2.1 If

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \tag{2.1}$$

show that

$$c_k = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi k f_0 t}\, dt \tag{2.2}$$

2.2 Find c_k for (1.1). In particular, account for the fact that the signal has period $1/(2f_0)$, although the complex series is written using the underlying frequency f_0.

2.3 Verify (1.1) using Python.

2.4 Show that

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \left(a_k \cos(2\pi k f_0 t) + b_k \sin(2\pi k f_0 t)\right) \tag{2.3}$$

and obtain the coefficients

$$a_0 = 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\, dt,$$

$$a_k = 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) \cos(2\pi k f_0 t)\, dt,$$

$$b_k = 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) \sin(2\pi k f_0 t)\, dt, \quad k$$

$geq1$.

2.5 Find a_k and b_k for (1.1). Use the same coefficient convention as in the real Fourier series above.

2.6 Verify (2.3) using Python.

For this signal, we can check the coefficients using these results: $b_k = 0$ for every k, $a_0/2 = 2A_0/\pi$, and $a_{2n} = -4A_0/[\pi(4n^2 - 1)]$ for $n \geq 1$, while $a_k = 0$ for odd k. Equivalently, $c_0 = 2A_0/\pi$, $c_{2n} = c_{-2n} = -2A_0/[\pi(4n^2 - 1)]$, and all odd-indexed complex coefficients vanish.

3. Fourier Transform

3.1 The Dirac delta is a generalized function characterized by

$$\delta(t) = 0, \quad t \neq 0, \tag{3.1}$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1, \tag{3.2}$$

$$\int_{-\infty}^{\infty} g(t)\delta(t - t_0)\, dt = g(t_0). \tag{3.3}$$

3.2 The Fourier Transform of $g(t)$ is

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t}\, dt. \tag{3.4}$$

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3.3 Show that

$$g(t - t_0) \xleftrightarrow{\mathcal{F}} G(f)e^{-j2\pi f t_0}. \tag{3.5}$$

3.4 Show that

$$G(t) \xleftrightarrow{\mathcal{F}} g(-f). \tag{3.6}$$

3.5 Determine the Fourier transform of $\delta(t)$, treating it as a generalized function (distribution).
3.6 Determine the Fourier transform of $e^{-j2\pi f_0 t}$.
3.7 Determine the Fourier transform of $\cos(2\pi f_0 t)$.
3.8 Find the Fourier transform of the periodic signal $x(t)$ and plot its line spectrum. Use the distributional Fourier-transform convention and verify the result using Python.
3.9 Using the normalized definitions $\mathrm{rect}\,(t) = 1$ for $|t| < 1/2$ and $\mathrm{sinc}\,(t) = \sin(\pi t)/(\pi t)$, show that

$$\mathrm{rect}\,(t) \xleftrightarrow{\mathcal{F}} \mathrm{sinc}\,(f)\,. \tag{3.7}$$

Verify using Python.
3.10 Determine the Fourier transform of $\mathrm{sinc}\,(t)$ using the same normalized convention. Verify using Python.

4. Filter

4.1 Determine the required DC gain and the harmonic-rejection condition on $H(f)$ so that the steady-state output is a constant 5 V. State any assumptions about the filter passband.
4.2 For an ideal low-pass model with cutoff f_c satisfying $0 < f_c < 100\,\mathrm{Hz}$, find the corresponding impulse response $h(t)$.
4.3 Verify your result using convolution.

The input's average value is $X_{\mathrm{DC}} = 2A_0/\pi$. For $A_0 = 12\,\mathrm{V}$, the required DC gain is therefore $K = 5/(24/\pi) = 5\pi/24 \approx 0.6545$. An ideal low-pass response with $H(f) = K$ for $|f| < f_c$ and $H(f) = 0$ for $|f| > f_c$, where $0 < f_c < 100\,\mathrm{Hz}$, passes the DC component and rejects the first and all higher harmonics. With the normalized sinc definition, its impulse response is $h(t) = 2Kf_c\mathrm{sinc}\,((2f_c t))$.

5. Filter Design

5.1 Design a Butterworth approximation to the specified low-pass response $H(f)$. State the filter order, cutoff frequency, and any design tolerance used.
5.2 Design a Chebyshev approximation to the specified low-pass response $H(f)$. State the filter order, passband ripple, and cutoff frequency.
5.3 Design a circuit for your Butterworth filter.
5.4 Design a circuit for your Chebyshev filter.

6. Circuit Input and Transformer

Our calculations use a 220 V RMS, 50 Hz primary supply and the measured 10.53 V RMS secondary. The transformer steps down the voltage and provides galvanic isolation.

The modified charger circuit is shown in Fig. 1.

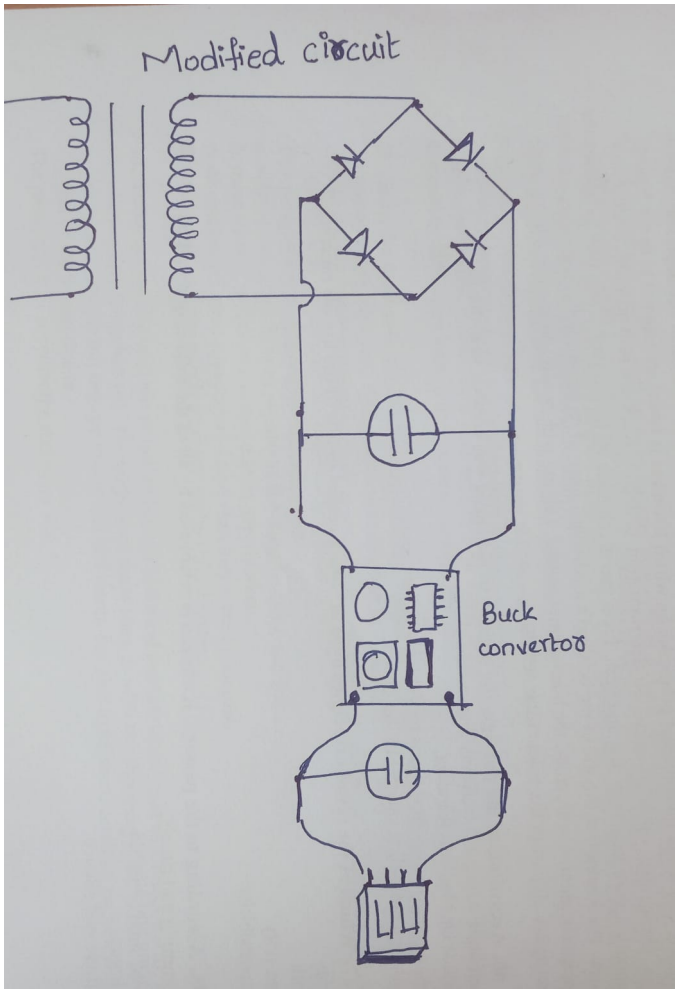


Fig. 1: Modified circuit schematic for the charger.

TABLE I: Input and transformer values

Quantity	Value
Primary RMS voltage	220 V
Mains frequency	50 Hz
Primary peak voltage (calculated)	311.1 V
Measured secondary RMS voltage	10.53 V
Secondary peak voltage (calculated)	14.89 V
Approximate turns ratio	20.9 : 1

We calculate peak voltage as $\sqrt{2}$ times RMS voltage. The ideal transformer voltage ratio also gives the turns ratio.

The separate centre-tapped example uses a 12–0–12 V transformer, a 12 V RMS winding output, and an approximately 18 V peak. These values do not apply to the 10.53 V RMS secondary used above.

7. Full-Wave Bridge Rectifier

In our four-diode bridge, two diodes conduct during each half-cycle, keeping current through the load in one direction and doubling the ripple frequency. Using a 0.7 V drop per diode, we calculate a 13.49 V peak, 8.59 V average, and 9.54 V RMS. These are first-order estimates; diode conduction and source impedance affect the actual waveform.

TABLE II: Bridge rectifier values

Quantity	Value
Measured secondary RMS voltage	10.53 V
Secondary peak voltage	14.89 V
Assumed forward drop per diode	0.7 V
Bridge peak voltage	13.49 V
Rectified average voltage	8.59 V
Rectified RMS voltage	9.54 V
Input frequency	50 Hz
Full-wave ripple frequency	100 Hz
Ripple period	10 ms

Two rectifier diode options are the 1N4007 diode (1 A average forward-current class and 1000 V repetitive peak reverse-voltage rating) and the 1N5408 diode (3 A average forward-current class).

8. Smoothing Capacitor

The 2200 μF smoothing capacitor charges near each rectified peak and supplies the load between peaks, reducing output ripple.

For small ripple and nearly constant load current, we estimate peak-to-peak ripple as

$$V_{r(pp)} \approx \frac{I_L}{f_r C}.$$

At $f_r = 100$ Hz and $C = 2200\,\mu$F:

$$V_{r(pp)} \approx 4.545 I_L \text{ V},$$

where I_L is in amperes. For a 13.49 V peak, the approximate average is

$$V_{\mathrm{DC}} \approx 13.49 - 2.273 I_L \text{ V}.$$

These are calculated estimates, not measured values.

The separate centre-tapped example specifies a 100 μF filter capacitor and an approximately 18 V DC level. This differs from the 2200 μF calculation above.

9. Regulator and Buck Converter

We compare two 5 V regulator options: the 7805 linear regulator and the LM2596 step-down converter.

A. 7805 Linear Regulator

The 7805 provides a fixed 5 V output. TI lists about 2 V dropout at a 1 A test condition; the actual value depends on the device, load, and temperature.

Its approximate power dissipation is

$$P_D \approx (V_{\mathrm{IN}} - V_{\mathrm{OUT}}) I_{\mathrm{OUT}}.$$

The 7805 family has current limiting and thermal protection. Its approximately 2.4 A current-limit figure is not a continuous rating; operation above 1.5 A depends on the device and cooling.

B. LM2596 Buck Converter

The LM2596 family specifies a 4.5–40 V input range, up to 3 A output under stated conditions, and a 150 kHz oscillator. A module's usable current also depends on its components, board, and cooling.

For an ideal buck converter, the duty ratio is

$$D = \frac{V_{\mathrm{OUT}}}{V_{\mathrm{IN}}}.$$

A 12 V to 5 V conversion gives an ideal duty ratio of about 41.7%. This is a reference calculation, not our measured USB operating point.

TABLE III: Regulator comparison

Feature	7805	LM2596
Method	Linear	Switching buck
Output	Fixed 5 V	Fixed or adjustable
Main consideration	Headroom and heat	Ripple, layout, and heat
Key specifications	About 2 V dropout at stated 1 A test	4.5–40 V input; up to 3 A; 150 kHz

10. USB Output Port

The USB port provides power through VBUS and return through GND. D+ and D- carry USB 2.0 data and can identify legacy charging sources. Type-C advertises current through its CC pins; USB-PD also requires a compatible controller and negotiation.

TABLE IV: USB charging current values

Port or mode	Current value
USB 2.0 SDP	Up to 500 mA*
USB 3.x SDP	Up to 900 mA*
BC1.2 CDP	Up to 1.5 A
BC1.2 DCP	Up to 1.5 A
USB Type-C current advertisement	1.5 A or 3 A at 5 V

**These are nominal maximum current values under the applicable USB specification and configuration/enumeration conditions; they are not guaranteed by the connector alone.*

The table lists legacy proprietary D+/D- signatures. They are implementation-specific, not universal USB rules.

TABLE V: Example legacy D+/D- charging signatures

D+	D-	Reported current
2.0 V	2.0 V	Approx. 0.5 A
2.0 V	2.7 V	Approx. 1.0 A
2.7 V	2.0 V	Approx. 2.1 A
2.7 V	2.7 V	Approx. 2.4 A

11. Measured USB Output

We measured an average USB output of 4.94 V and 0.60 A using a voltmeter and ammeter. The instrument also displayed 16 mAh, which we list separately because mAh measures charge, not current.

TABLE VI: Measured USB output values

Parameter	Measured value
Average output voltage	4.94 V
Average output current	0.60 A
Power output	2.96 W